

AI531C — T08 · Learn Your First Kernel: Student Sheet

Tue 29 Sep, 2:00–3:00 PM · run: python3 T08_studio.py (+ 'support' / 'extension')

Your mission

Learn 3×3 kernels by least squares in two acts — deblurring (learning WINS) and denoising (learning TIES a tuned Gaussian) — and name the regime difference.

Experiment

Act 1 (deblur): input 28.5 → fixed sharpen 31.4 → LEARNED 32.3 dB (asserted win); inspect the learned centre-heavy $\pm$ pattern. Act 2 (denoise): tuned Gaussian 28.0 vs learned 28.0 — a TIE (asserted).

Vary / break it

Extension preview live if time allows: learned 5×5 deblur hits 34.5 dB. Vary the patch count 2k→20k in Act 1 and watch stability.

Discuss

Q1 · The learned denoise kernel's weights sum to ~ 1.0 . Why must they, and what would a sum of 1.3 do?

Q2 · The learned deblur kernel has large $\pm$ values ($\approx +19$ centre, -8 cross). Connect its shape to the L8 sharpen family.

Q3 · State one task from your lane where you expect the 'tie' regime, and one where you expect the 'win' regime.

Reflect

This is Unit III's thesis in one hour: learning relocates design effort — it pays where design is hard.

Fast? EXTENSION cell: learned 5×5 deblur (34.5 dB) — why does reach help deblurring more than denoising? Struggling? SUPPORT cell: Act 1 with the kernel provided; apply, score, compare with the fixed sharpen. Nothing is due after class — the hour is the work.